Pro DevOps Engineer

Complete Roadmap for Java Developers

Ubuntu · AWS Free Tier · Spring Boot Microservices

6 Months · 8 Phases · 26 Weeks · Hands-On Projects

26 Weeks	**8 Phases**	**20+ Projects**	**AWS Free Tier**	**Java/Spring Boot**

How to Use This Roadmap

Follow each phase in order. Every phase ends with a hands-on project you deploy on AWS Free Tier. Mark each checklist item complete before moving to the next. Expected commitment: 2–3 hours/day on weekdays. Total duration: ~26 weeks (6 months).

Table of Contents

PHASE 0 | **Linux & Networking Foundations** | *Week 1 – 2*

As a Java developer you already know how to code — now you need to master the OS and network layer that everything runs on. Ubuntu is your daily driver; these skills are prerequisite to every later phase.

Week 1 — Linux CLI Mastery

Core Commands

- File system navigation: ls, cd, pwd, find, locate
- File operations: cp, mv, rm, mkdir, chmod, chown, ln -s
- Text processing: cat, less, head, tail, grep, awk, sed, cut, sort, uniq
- Process management: ps, top, htop, kill, systemctl, journalctl
- Package management: apt update/upgrade, apt install/remove, dpkg
- Disk & memory: df, du, free, lsblk, mount
- Archiving: tar, zip, gzip, rsync
- User management: useradd, passwd, sudo, /etc/sudoers
- Shell scripting: variables, loops, if/else, functions, cron jobs

Week 1 Project

Shell Script Automation Suite

Write 5 bash scripts: (1) system health checker (CPU/RAM/Disk alerts), (2) log rotator with compression, (3) automated backup to a local folder, (4) user creation wizard, (5) Spring Boot app start/stop/status manager.

Week 2 — Networking & SSH

Networking Fundamentals

- OSI model layers and their DevOps relevance
- IP addressing, CIDR notation, subnets (e.g. 10.0.0.0/16)
- DNS: A, CNAME, MX records; dig, nslookup commands
- Ports & protocols: TCP/UDP, HTTP/HTTPS, SSH (22), common app ports
- Tools: ping, traceroute, netstat, ss, curl, wget, nc (netcat)
- Firewall: ufw enable/disable, ufw allow/deny, iptables basics
- SSH: key pair generation, ssh-keygen, ~/.ssh/config, scp, sftp
- VPN concepts: tunneling, port forwarding, ssh -L/-R/-D
- Load balancing concepts: round-robin, least-connections
- TLS/SSL: certificates, openssl commands, Let's Encrypt basics

Week 2 Project

Secure Multi-Server Setup

On your Ubuntu machine: run two Java apps on ports 8080 and 8081. Configure ufw to allow only those ports. Set up SSH key-based login with a config alias. Use nginx as a local reverse proxy routing /app1 → 8080 and /app2 → 8081.

PHASE 1 **Git, Docker & Containerisation** *Week 3 – 5*

Week 3 — Git Advanced

- Branching strategies: GitFlow, trunk-based development, GitHub Flow
- git rebase vs merge — when to use each
- Interactive rebase: squash, fixup, reorder commits
- git stash, git cherry-pick, git bisect
- Resolving merge conflicts like a pro
- git hooks: pre-commit (lint/test), commit-msg (conventional commits)
- GitHub: PRs, code reviews, branch protection rules, GitHub Actions triggers
- Signed commits with GPG keys
- Monorepo vs polyrepo patterns for microservices
- Git submodules and subtrees

Week 4 — Docker

- Docker architecture: daemon, client, registry, image, container, layer
- Dockerfile: FROM, RUN, COPY, WORKDIR, ENV, EXPOSE, CMD, ENTRYPOINT
- Multi-stage builds for Java — build with Maven, run with JRE only
- docker build, run, exec, logs, inspect, stats, rm, rmi
- Docker networking: bridge, host, overlay — container-to-container comms
- Docker volumes: bind mounts vs named volumes
- Docker Hub: push/pull, private repos, image tagging strategy
- docker compose: services, networks, volumes, depends_on, env_file
- .dockerignore — exclude target/ and .git from build context
- Docker security: non-root user, read-only filesystem, image scanning

Week 4 — Spring Boot Docker

- Spring Boot Maven/Gradle Docker plugin (Jib) — build without Dockerfile
- Layered JARs for optimal Docker cache efficiency
- Environment-specific config via ENV / Spring profiles
- Health checks in Dockerfile and compose
- Connecting Spring Boot to MySQL/Postgres containers via compose

Week 5 — Docker Compose Full Stack

- Run multi-service Spring Boot app: API + DB + Redis + nginx
- Override files: docker-compose.override.yml for local dev
- Secrets management in compose (file-based)
- Resource limits: mem_limit, cpu_shares
- Logging drivers: json-file, syslog

Phase 1 Project

Containerised Spring Boot Microservices

Build 3 Spring Boot services (User, Order, Product). Write optimised multi-stage Dockerfiles. Wire them with docker-compose including MySQL, Redis, and nginx reverse proxy. Push images to Docker Hub. Document the Compose setup.

PHASE 2 **CI/CD Pipelines** *Week 6 – 8*

Week 6 — GitHub Actions

- Workflow YAML: on, jobs, steps, uses, env, secrets
- Runners: ubuntu-latest, self-hosted runners on your Ubuntu machine
- Build & test Spring Boot: actions/checkout, setup-java, mvn test
- Cache Maven dependencies: actions/cache
- Build and push Docker image: docker/build-push-action
- Matrix builds: test across Java 17/21
- Environment secrets: DOCKER_HUB_TOKEN, AWS_ACCESS_KEY_ID
- Artifacts: upload-artifact, download-artifact
- Deployment jobs with needs: and environment: approvals
- Reusable workflows: workflow_call

Week 7 — Jenkins

- Install Jenkins on Ubuntu via Docker
- Jenkinsfile: declarative pipeline syntax
- Stages: Checkout → Build → Test → Docker Build → Push → Deploy
- Jenkins agents and parallel stages
- Credentials management in Jenkins (not plaintext)
- Blue Ocean UI for pipeline visualisation
- Webhooks: trigger Jenkins from GitHub push
- SonarQube integration for code quality gates
- Nexus/Artifactory for Maven artifact storage
- Pipeline shared libraries

Week 8 — Quality & Security Gates

- Unit tests: JUnit 5, Mockito — coverage report with JaCoCo
- Integration tests: Testcontainers (real DB in pipeline)
- Code quality: SonarQube quality gate (block merge if fails)
- Dependency scanning: OWASP Dependency-Check Maven plugin
- Docker image scanning: Trivy in CI pipeline
- Conventional commits + semantic versioning automation
- Changelog generation: git-cliff or standard-version
- Slack/email notifications on build failure

Phase 2 Project

Full CI/CD Pipeline

GitHub Actions pipeline: on PR → run tests + SonarQube + Trivy scan. On merge to main → build Docker image → push to ECR → deploy to EC2 via SSH. Jenkins pipeline as alternative with same stages. Enforce quality gate: PRs blocked if coverage < 80%.

PHASE 3

AWS Core Services

Week 9 – 12

Week 9 — Identity & Networking (IAM + VPC)

- IAM: users, groups, roles, policies (inline vs managed)
- Least-privilege principle — deny by default
- IAM roles for EC2, Lambda, ECS tasks (no hardcoded keys!)
- STS: AssumeRole, temporary credentials
- MFA enforcement, password policy, IAM Access Analyzer
- VPC: CIDR, subnets (public/private), route tables, internet gateway
- NAT Gateway vs NAT Instance (cost trade-off on Free Tier)
- Security Groups vs NACLs — stateful vs stateless
- VPC Peering and VPC Endpoints (S3/DynamoDB gateway endpoints)
- AWS CLI: configure, profiles, --query, --output json/table

Week 10 — Compute (EC2 + Auto Scaling + ALB)

- EC2: instance types, AMIs, key pairs, user data scripts
- Free Tier: t2.micro/t3.micro — one always-free instance
- EC2 Spot instances for CI runners (80% cheaper)
- Launch Templates and Auto Scaling Groups
- ALB: listeners, target groups, health checks, path-based routing
- Deploy Spring Boot on EC2: systemd service, auto-start on reboot
- Elastic IP vs public IP — when to use each
- EC2 Instance Connect and SSM Session Manager (no SSH port needed)
- AMI creation from running instance (backup strategy)
- EC2 cost optimisation: stop when not in use, Savings Plans

Week 11 — Storage & Database (S3 + RDS)

- S3: buckets, objects, prefixes, storage classes
- S3 versioning, lifecycle rules, replication
- S3 static website hosting, pre-signed URLs
- S3 SDK in Spring Boot: upload, download, delete, list objects
- S3 event notifications → Lambda/SQS
- RDS: MySQL/PostgreSQL on Free Tier (db.t3.micro, 20GB)
- RDS Multi-AZ vs read replicas
- RDS automated backups, snapshots, point-in-time recovery
- Connect Spring Boot to RDS (spring.datasource in Secrets Manager)
- ElastiCache: Redis on Free Tier — Spring Boot cache integration

Week 12 — Containers on AWS (ECR + ECS)

- ECR: create repo, push/pull Docker images, lifecycle policies
- ECS concepts: cluster, task definition, service, container
- ECS launch types: Fargate (serverless) vs EC2
- Fargate Free Tier: limited — use EC2 launch type to stay free
- Task definitions: CPU/memory, env vars, secrets from SSM/SecretsManager

- ECS service: desired count, rolling update deployment
- ECS with ALB: service discovery via target groups
- ECS Exec — exec into running container (like docker exec)
- CloudWatch Logs for ECS container output
- ECS autoscaling: Application Auto Scaling on CPU/memory

Phase 3 Project

Production-Grade AWS Deployment

Deploy the Phase 1 microservices to AWS: VPC with public/private subnets. EC2 instances in private subnet behind ALB. RDS MySQL in private subnet. S3 for file storage integrated into User service. CI/CD pipeline pushes to ECR and triggers ECS rolling deployment. All secrets in AWS Secrets Manager.

Week 13 — Kubernetes Fundamentals (Local)

- Install minikube or kind on Ubuntu for local K8s
- Core objects: Pod, ReplicaSet, Deployment, Service, ConfigMap, Secret
- Namespaces: separate dev/staging/prod workloads
- kubectl: get, describe, apply, delete, logs, exec, port-forward
- YAML manifests: write from scratch — no GUI copy-paste
- Deployments: rolling update, rollback (kubectl rollout undo)
- Services: ClusterIP, NodePort, LoadBalancer types
- Persistent Volumes and Persistent Volume Claims
- Resource requests and limits (cpu/memory) — always set these!
- Liveness and readiness probes for Spring Boot (/actuator/health)

Week 14 — Kubernetes Intermediate

- Ingress controllers: nginx-ingress, path/host-based routing
- ConfigMaps and Secrets — mount as env vars or volume files
- Horizontal Pod Autoscaler (HPA) based on CPU
- StatefulSets for databases (Postgres on K8s)
- DaemonSets: node-level agents (log collectors, monitoring)
- Jobs and CronJobs for batch tasks
- RBAC: ClusterRole, Role, RoleBinding — least privilege in K8s
- Network Policies: restrict pod-to-pod communication
- Pod Disruption Budgets for zero-downtime maintenance
- Kustomize: environment-specific overlays without Helm

Week 15 — Helm

- Helm concepts: chart, release, values.yaml, templates
- helm install, upgrade, rollback, uninstall, list
- Write a Helm chart for your Spring Boot service
- values.yaml overrides per environment (values-prod.yaml)
- Helm hooks: pre-install database migrations
- Artifact Hub: find charts for nginx-ingress, cert-manager, prometheus
- Helmfile: manage multiple charts declaratively
- Chart testing with helm lint and helm template

Week 16 — Amazon EKS

- EKS: managed control plane, you manage node groups
- eksctl: create cluster, add node group, delete cluster
- Free Tier warning: EKS control plane costs $0.10/hour — spin up only when practising
- IAM roles for service accounts (IRSA) — pods assume AWS IAM roles
- AWS Load Balancer Controller — replaces in-tree ALB integration
- EBS CSI driver for persistent storage on EKS
- EKS managed add-ons: CoreDNS, kube-proxy, VPC CNI

- Cluster Autoscaler or Karpenter for node autoscaling
- kubectl auth: aws eks update-kubeconfig
- Deploy Phase 1 microservices to EKS with Helm charts

Phase 4 Project

Kubernetes Microservices Deployment

Write K8s manifests + Helm charts for all 3 microservices. Deploy to minikube locally. Configure HPA, readiness probes, resource limits. Set up nginx Ingress with path routing. Deploy to EKS (spend 2–3 hours max, then delete cluster to save cost). CI/CD: helm upgrade --install in GitHub Actions.

PHASE 5

Infrastructure as Code (IaC)

Week 17 – 19

Week 17 — Terraform

- Terraform concepts: provider, resource, data source, output, variable
- HCL syntax: blocks, arguments, expressions, functions
- terraform init, plan, apply, destroy, state
- State management: local vs S3 backend with DynamoDB locking
- AWS provider: configure with IAM role (not access keys in code!)
- Resources: aws_vpc, aws_subnet, aws_security_group, aws_instance
- Resources: aws_s3_bucket, aws_rds_instance, aws_ecs_service
- Modules: write reusable VPC, EC2, RDS modules
- Variables: terraform.tfvars, environment variable TF_VAR_*
- Workspace: terraform workspace for dev/staging/prod

Week 18 — Terraform Advanced + CloudFormation

- Remote state: S3 backend, state locking with DynamoDB
- terraform import: bring existing resources under IaC management
- Terraform Cloud (free tier) for state and remote runs
- for_each and count for creating multiple similar resources
- Dynamic blocks and conditional expressions
- Terraform modules from Terraform Registry
- CloudFormation basics: stacks, templates, parameters, outputs
- CloudFormation vs Terraform — when AWS uses CFN natively (CDK, SAM)
- AWS CDK: write infrastructure in Java — create a CDK app
- CDK constructs: L1 (cfn), L2 (opinionated), L3 (patterns)

Week 19 — Ansible

- Ansible concepts: inventory, playbook, task, module, role
- Install and configure software on EC2 via Ansible
- Ansible roles: structure, defaults, vars, tasks, handlers
- Ansible Vault: encrypt secrets in playbooks
- Ansible Galaxy: download community roles
- Use Ansible to configure EC2 after Terraform provisions it
- Idempotency: run playbooks multiple times safely
- Dynamic inventory: aws_ec2 plugin auto-discovers instances

Phase 5 Project

Full IaC-Managed AWS Environment

Terraform code that provisions: VPC, subnets, SGs, ALB, EC2 ASG, RDS, ElastiCache, S3, ECR. Remote state in S3 + DynamoDB. Modules for VPC and compute. Ansible playbook configures EC2 (Java, Docker, nginx). CDK bonus: rewrite VPC + ECS in Java CDK.

PHASE 6 **Monitoring & Observability** *Week 20 – 21*

Week 20 — AWS CloudWatch + Prometheus + Grafana

- CloudWatch metrics: EC2, RDS, ECS, ALB built-in metrics
- CloudWatch Logs: log groups, log streams, log insights queries
- CloudWatch Alarms: SNS notifications for CPU > 80%, error rates
- CloudWatch Dashboards: composite view of your stack
- CloudWatch Agent on EC2: collect custom metrics and logs
- AWS X-Ray: distributed tracing for Spring Boot (add Spring Cloud Sleuth + X-Ray)
- Prometheus: install on EC2, scrape Spring Boot /actuator/prometheus
- Spring Boot Actuator: expose metrics with Micrometer + Prometheus registry
- Grafana: connect to Prometheus, import Spring Boot dashboard (ID 12900)
- Grafana alerts: PagerDuty/Slack integration

Week 21 — ELK Stack + Distributed Tracing

- ELK: Elasticsearch + Logstash + Kibana — run via Docker Compose
- Filebeat: ship Spring Boot logs from EC2 to Elasticsearch
- Logback JSON format in Spring Boot for structured logging
- Kibana: create index patterns, visualizations, dashboards
- OpenTelemetry: instrument Spring Boot with OTLP exporter
- Jaeger: distributed tracing backend — run locally with Docker
- Trace context propagation across microservices (W3C TraceContext)
- SLI/SLO/SLA definitions for your services
- Error budget concept and alerting strategy
- Runbook template: how to respond to each alert

Phase 6 Project

Full Observability Stack

Spring Boot services expose metrics + traces + structured logs. Prometheus scrapes metrics → Grafana dashboard with 5 key panels. ELK stack receives all logs → Kibana error dashboard. Jaeger shows end-to-end trace for cross-service requests. CloudWatch alarm triggers SNS email when error rate > 5%.

PHASE 7

Security & DevSecOps

Week 22 – 23

Week 22 — AWS Security Services

- AWS Secrets Manager: store DB passwords, API keys — rotate automatically
- AWS KMS: create CMKs, encrypt S3, RDS, EBS — envelope encryption
- AWS Cognito: user pools (auth), identity pools (AWS resource access)
- Spring Boot + Cognito: JWT validation, OAuth2 resource server
- AWS WAF: web ACL, rate limiting, SQL injection rules on ALB/CloudFront
- AWS Shield Standard (free): DDoS protection
- AWS GuardDuty: threat detection (30-day free trial)
- AWS Config: track configuration changes, compliance rules
- AWS Security Hub: aggregated security findings
- AWS Inspector: vulnerability scanning for EC2 and ECR images

Week 23 — DevSecOps in CI/CD

- SAST: SpotBugs + SonarQube in Maven build — block on HIGH issues
- DAST: OWASP ZAP automated scan against deployed app
- SCA: OWASP Dependency-Check — CVE scanning of Maven dependencies
- Secret scanning: git-secrets / truffleHog — prevent keys in git
- Docker image scanning: Trivy in CI, fail on CRITICAL CVEs
- Kubernetes security: Pod Security Standards, OPA Gatekeeper
- Network segmentation: SGs allow only required ports, NACLs as backstop
- Principle of least privilege in IAM — audit with IAM Access Analyzer
- Compliance as code: AWS Config rules checked in pipeline
- OWASP Top 10 — map each to a Spring Boot mitigation

Phase 7 Project

Secure Spring Boot Deployment

Cognito-protected API Gateway → Spring Boot (JWT validation). All secrets in Secrets Manager. KMS-encrypted S3 bucket and RDS. WAF rules on ALB. CI pipeline: SAST + SCA + Trivy scan before deploy. GuardDuty enabled. IAM roles with minimal permissions audited by Access Analyzer.

PHASE 8 — **Advanced DevOps & Certifications** — *Week 24 – 26*

Week 24 — Advanced Deployment Strategies

- Blue/Green deployment on ECS and EKS — zero downtime
- Canary deployment: Route 53 weighted routing or AWS CodeDeploy
- Feature flags: AWS AppConfig or LaunchDarkly
- GitOps with ArgoCD: deploy K8s manifests from git automatically
- Flux CD: alternative GitOps operator
- Rollback automation: auto-rollback on CloudWatch alarm
- Database migrations in CI/CD: Flyway/Liquibase with zero-downtime patterns
- AWS CodePipeline + CodeDeploy: native AWS CI/CD alternative

Week 25 — Cost Optimisation & SRE Practices

- AWS Cost Explorer: analyse spending by service/tag
- Resource tagging strategy: env, team, project tags on all resources
- AWS Compute Optimizer: rightsizing recommendations
- Spot Instances for batch jobs and CI runners (save 70–90%)
- Reserved Instances and Savings Plans for stable workloads
- S3 Intelligent-Tiering for variable-access objects
- SRE concepts: error budgets, toil reduction, blameless postmortems
- Chaos engineering: AWS Fault Injection Simulator (FIS)
- Runbooks and incident response playbooks
- On-call rotations and escalation policies (PagerDuty)

Week 26 — Certifications Roadmap

AWS CCP	AWS Certified Cloud Practitioner — broad overview, good if you want a starting cert.	*~1 month prep*
AWS SAA-C03	AWS Solutions Architect Associate — THE key cert for DevOps on AWS. Study this first.	*~2 months prep*
AWS DVA-C02	AWS Developer Associate — covers CodePipeline, Lambda, DynamoDB deeply.	*~1.5 months prep*
AWS SysOps	AWS SysOps Administrator — operations, monitoring, automation.	*~2 months prep*
AWS DevOps Pro	AWS DevOps Engineer Professional — gold standard. Requires SAA or DVA first.	*~3 months prep*
CKA	Certified Kubernetes Administrator — essential for K8s-heavy roles.	*~2 months prep*
Terraform Assoc	HashiCorp Terraform Associate — validates IaC skills.	*~3 weeks prep*

Recommended Cert Order:

- Month 7: AWS Solutions Architect Associate (SAA-C03)
- Month 9: CKA — Certified Kubernetes Administrator
- Month 11: Terraform Associate
- Month 13: AWS DevOps Engineer Professional

Service	Free Tier Allowance	Expires
EC2 t2.micro	750 hrs/month	12 months
RDS db.t3.micro	750 hrs/month + 20GB storage	12 months
S3	5GB storage, 20K GET, 2K PUT	Forever
Lambda	1M requests/month, 400K GB-sec	Forever
CloudWatch	10 custom metrics, 10 alarms, 5GB logs	Forever
DynamoDB	25GB storage, 25 RCU/WCU	Forever
ECR	500MB/month private storage	Forever
SNS	1M publishes, 100K HTTP deliveries	Forever
SQS	1M requests/month	Forever
Secrets Manager	30-day free trial per secret	Trial only
EKS	$0.10/hour for control plane — NOT free!	—
NAT Gateway	NOT free — use NAT Instance (t2.micro)	—

Essential Learning Resources

Resource	What It Covers	Cost
AWS Skill Builder	Official AWS courses, practice exams, labs	Free tier available
Linux Journey (linuxjourney.com)	Interactive Linux CLI learning	Free
Play with Docker (labs.play-with-docker.com)	Browser-based Docker labs	Free
Killercoda	K8s, Linux, Docker labs in browser	Free
A Cloud Guru / Udemy	AWS + DevOps video courses	Paid (~$15)
Terraform Up & Running (book)	Terraform best practices by Yevgeniy Brikman	Paid
The DevOps Handbook (book)	DevOps culture and practices	Paid
AWS Well-Architected Framework docs	Official AWS architecture pillars	Free
GitHub: awesome-devops	Curated DevOps tools and resources	Free

26-Week Quick Reference Timeline

Wk 1 — Linux CLI — shell scripting, file system, processes	**Wk 2** — Networking — TCP/IP, DNS, SSH, ufw, nginx reverse proxy
Wk 3 — Git advanced — branching, rebase, hooks, GPG signing	**Wk 4** — Docker — Dockerfiles, multi-stage, compose, Spring Boot

Wk 5 — Docker Compose full stack — multi-service, volumes, secrets	**Wk 6** — GitHub Actions — build/test/push pipeline, matrix builds
Wk 7 — Jenkins — Jenkinsfile, webhooks, SonarQube, Nexus	**Wk 8** — Quality gates — JaCoCo, Testcontainers, Trivy, OWASP DC
Wk 9 — AWS IAM + VPC — least privilege, subnets, SGs, NACLs	**Wk 10** — EC2 + ALB + Auto Scaling — deploy Spring Boot on AWS
Wk 11 — S3 + RDS + ElastiCache — storage and database on AWS	**Wk 12** — ECR + ECS — containerised deployment with rolling updates
Wk 13 — Kubernetes fundamentals — pods, deployments, services	**Wk 14** — K8s intermediate — Ingress, RBAC, HPA, NetworkPolicy
Wk 15 — Helm — write charts, upgrade, rollback, helmfile	**Wk 16** — EKS — managed K8s on AWS, IRSA, ALB controller
Wk 17 — Terraform — HCL, state, modules, AWS resources	**Wk 18** — Terraform advanced + CloudFormation + AWS CDK (Java)
Wk 19 — Ansible — playbooks, roles, dynamic inventory, vault	**Wk 20** — CloudWatch + Prometheus + Grafana — metrics and dashboards
Wk 21 — ELK + OpenTelemetry + Jaeger — logs and distributed tracing	**Wk 22** — AWS security — Cognito, KMS, Secrets Manager, WAF, GuardDuty
Wk 23 — DevSecOps — SAST, DAST, SCA, secret scanning in CI/CD	**Wk 24** — Blue/Green, canary, GitOps (ArgoCD), Flyway migrations
Wk 25 — Cost optimisation, SRE, chaos engineering, FIS	**Wk 26** — Certification prep — SAA-C03, CKA, Terraform Associate

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Generated for Bulbul Ahmed · 6+ Years Java/Spring Boot · AWS Free Tier · Ubuntu · June 2026 · Follow phases in order — each builds on the last. Good luck! ■